



LISTENING

#11

1 - You will hear a weather report from a local TV station. Listen carefully and answer the following questions about the forecast and the meteorologist's advice.

The Weather Forecast.

1 - According to the weather reporter, what should you do if you want to take your dog for a walk?

2 - What weather conditions are currently happening in Basin?

3 - Until when are the flash flood warnings in effect?

4 - What is the forecast for the overnight hours?

5 - When are conditions expected to start drying out?

6 - At what time does the weather reporter recommend running errands?

7 - In which areas will the temperature range between the 60s and 70s?

8 - What is the forecast for Cedar City, Saint George, and Moab tomorrow?

9 - What does the reporter mean when saying, "Please stay weather aware"?

10 - Why should people plan to get out early next week?

11 - When is the air quality expected to improve?

12 - Why is tomorrow unhealthy for sensitive groups?

How Reliable Are Weather Forecasts?

The Short Answer:

A seven-day forecast can accurately predict the weather about 80 percent of the time and a five-day forecast can accurately predict the weather approximately 90 percent of the time. However, a 10-day—or longer—forecast is only right about half the time.

If you want to know what the weather will be like within the next week, a weather forecast can give you a really good idea of what to expect. A seven-day forecast can accurately predict the weather about 80 percent of the time and a five-day forecast can accurately predict the weather approximately 90 percent of the time.

However, a 10-day—or longer—forecast is only right about half the time. Meteorologists use computer programs called weather models to make forecasts. Since we can't collect data from the future, models have to use estimates and assumptions to predict future weather. The atmosphere is changing all the time, so those estimates are less reliable the further you get into the future.

How Weather Forecasts Are Made

Some of the information needed to make a weather forecast comes from environmental satellites. NOAA, the National Oceanic and Atmospheric Administration, operates three types of environmental satellites that monitor Earth's weather:

Geostationary satellites: NOAA's [Geostationary Environmental Operational Satellite-R](#) (GOES-R) series satellites orbit approximately 22,000 miles above Earth and they provide a picture of what the weather is like right now. "Geostationary" means that the satellites orbit at the same rate that the Earth rotates. This means they can collect near-continuous images over the same area. Because they focus on one spot, they can provide up-to-the-minute information about severe weather. This information helps forecasters understand how quickly a storm, such as a hurricane, is growing and moving.

Polar-orbiting satellites: Satellites as part of NOAA's [Joint Polar Satellite System](#) (JPSS) orbit approximately 500 miles above Earth. They zip around our planet from pole to pole 14 times per day. Because they orbit while the Earth is rotating below, these satellites can see every part of Earth twice each day. Polar orbiting satellites can monitor the entire Earth's atmosphere, clouds and oceans at high resolution. By watching these global weather patterns, polar orbiting satellites can help meteorologists accurately predict long-term forecasts—up to 7 days in the future.

Deep space satellite: NOAA's Deep Space Climate Observatory (DSCOVR) orbits one million miles from Earth. It provides space weather alerts and forecasts while also monitoring the amounts of solar energy absorbed by Earth every day. DSCOVR also makes observations about ozone and aerosols in Earth's atmosphere. These factors are important in making air quality forecasts.

Polar orbiting satellites provide the information most useful for long-term weather forecasting. These satellites use instruments to measure energy, called radiation, emitted by the Earth and atmosphere. This information is incorporated into weather models, which in turn leads to more accurate weather forecasts. Other instruments can also be used to map sea surface temperature—an important factor in long-term weather forecasting.

The satellite performs these accurate measurements all around the globe twice per day. This flood of data is what helps weather forecasters to reliably predict the weather up to 7 days in advance. These measurements can also help forecasters predict seasonal weather patterns, such as [El Niño](#) and [La Niña](#).

Polar orbiting satellites collect essential information for the models that forecast severe weather like hurricanes, tornadoes and blizzards days in advance. The information they collect is also needed to assess environmental hazards such as droughts, forest fires, poor air quality, and harmful coastal waters.

<https://scijinks.gov/forecast-reliability/>

COMPREHENSION QUESTIONS

1 – Why are weather forecasts for 7 days or fewer more reliable than forecasts of 10 days or longer?

2 – What role do polar-orbiting satellites play in improving the accuracy of long-term weather forecasts?